**MDC Subjects of**
**Computer Science and Applications**

# Paper Title:E-Commerce
## Paper Code:CSA-MDC-101/201/301

| Th | P | Cr | |
|---|---|---|---|
| 3 | - | 3 | |

**TheoryExternalMarks:75**
**Theory Internal Marks:--**
**Level: L-100**
**Time Duration:3Hrs.** **NumberofTheoryHours:45**

**Objectives: The objective of this course is to understand the process of electronic commerce and familiarizes students with the technology involved in it.**

## Learning Outcomes:

**After completing the course students will be able to**

i) Know about the Basic of E-commerce, its type, EDI, its structure and components
ii) Understand the basic of Web Based E-Commerce, Web Site Development and online promotion tools and techniques.
iii) Gainanin sight into electronic payment systems, mobile commerce and various application of e-commerce along with some case studies.

## Note:

i. The Question Paper will consist of Four Sections.
ii. Examinerwillsettotalof**NINE**questionscomprising**TWO**questionsfromeachSectionand **ONE compulsory question of short answer type covering whole syllabi.**
iii. Thestudentsarerequiredtoattempt**ONE**questionfromeachSectionandtheCompulsory question.
iv. All questions carry equal marks unless specified.

## UNIT-I

**An OverviewofE-Commerce:**

**Definitions:E-commerce,E-business,differencebetweenE-commerceandE-business,Problemswith Traditional business systems, Aims of E-commerce, Types of E-commerce: B2B, B2C, C2C, B2G, G2H, G2C, Operational & Strategic benefits of E-commerce, Issues & Challenges in E-commerce. Electronic Data Interchange (EDI): Definition; Traditional versus EDI enabled system for document exchange;ComponentsofEDI:EDIStandards,EDISoftware,CommunicationNetworks;EDIMessage Structure; EDI Notification Structure; EDIin India; EDIenabled procurement process; Benefits of EDI: Direct Benefits, Strategic Benefits; EDI Implementation issues; Legal Aspects**

## UNIT-II

**Web based E-Commerce: Definition; Need for web based business, Steps in setting up business on Internet:Selection&registrationofdomainname,Websitedevelopment:Planningawebsite,Stepsfor creating a website, Elements of a webpage, web authoring tools, Hosting a website:Website hosting considerations.**
**OnlinePromotiontools&techniques:Gettinglinkstoyoursite,banneradvertisements&measuring advertisement effectiveness; Web Traffic Analysis: Hits, View pages, Visits and Other web-reporting tools, various measures, What is Search Engine optimization**

## UNIT-III

**ElectronicPaymentSystems:E-cash:Purchasing&usingofe-cash;ElectronicPursestheirloading with cash and use; E-cheque payment system; Online Third Party Verified Payment System through Credit & Debit Cards; ATM based cash disbursement system; Electronic Bill Payment System; Inter bank clearing system.**

## UNIT-IV

**Mobile Commerce: Definition, Benefits of Mobile Commerce, Issues in Mobile Commerce, Mobile Commerce Framework**

**Applications of E-Commerce & Case Studies**: Applications of e-commerce, Case studies in Retailing, Banking and e-governance; Cyber Crimes: Types, Cyber Forensics, Cyber crimes and IT Act - 2000.

**SuggestedReadings:**


1. Bhasker,Bharat: ElectronicCommerce:Framework,TechnologiesandApplications;Tata McGraw- Hill.
2. Bajaj,Kamlesh&Nag,Debjani:E-Commerce-TheCuttingEdgeofBusiness;TataMcGraw Hill.
3. Young,MargaretLevine:TheCompleteReference:Internet;TataMcGrawHill.
4. KalaKota,Ravi&Whinston,AndrewB.:FrontiersofElectronicCommerce;AddisonWesley.
5. Stallings,William:NetworkSecurityEssentials:Applications&Standards;PearsonEducation.
6. Minoli,Daniel&Minoli,Emma:WebCommerceTechnologyHandbook;TataMcGrawHill.
7. MurthyCSV:e-Commerce:Concepts,Models,Strategies;HimalayasPublishingHouse.
8. Kosiur,David:UnderstandingE-Commerce;MicrosoftPress.

# Paper Title: Management Information System

## PaperCode:BCA-MDC-102/202/302

| Th | Pr | Cr | TheoryExternalMarks:75 |
|---|---|---|---|
| 3 | - | 3 | TheoryInternalMarks:-- |

**Level: L-100**

**TimeDuration:3Hrs.** **NumberofTheoryHours:45**

### Objectives:

1. To understand and asses the importance of information and its role in business.
2. To develop data analyzing skills in students to evaluate information and the tools used for information processing.
3. To imbibe theoretical knowledge of MIS in the students and prepare the students technological competitive and make them ready to self-upgrade with the higher technical skill.

### Learning Outcomes:

1. Understand the information needs of an organization and a business function.
2. Evaluate effectiveness of decision making process and identify its tools
3. Understand DSS techniques for making effective decisions
4. Design parameters for MIS application, for data analysis uses

***Note:*** Examiner will be required to set nine questions in all. First question will be compulsory, consisting of objective type/short answer type questions covering the entire syllabus. In addition to that eight more questions will be set, two questions from each unit. A candidate will be required to answer five questions in all, selecting one question from each unit in addition to compulsory question number one. All questions will carry equal marks**.**

## UNIT-I

Introduction to System and Basic System Concepts, Types of Systems, The Systems Approach, Information System: Definition & Characteristics, Types of information, Role of Information in Decision-Making.

## UNIT-II

**An overview of Management Information System:** Definition & Characteristics, Components of MIS, Frame Work for Understanding MIS: Information requirements & Levels of Management, Simon's Model of decision-Making, Structured Vs Unstructured decisions, Formal vs. Informal Systems**.**

## UNIT-III

**Developing Information Systems:** Analysis & Design of Information Systems, Implementation & Evaluation,Pitfalls in MIS Development**.**

## UNIT-IV

**Functional MIS: A Study of Personnel, Financial and Production MIS, Introduction to E Business Systems, E-Commerce- Technologies, Applications, Decision Support Systems Support Systems for Planning, Control and Decision- Making.**

**TEXTANDREFERENCEBOOKS**:


1. J.Kanter,"Management/InformationSystems",PHI
2. GordonB.Davis,M.H.Olson,"ManagementInformationSystems-ConceptualFoundations,Structure and Development", McGraw Hill.
3. JamesA.O'brien,"ManagementInformationSystems",TataMcGrawHill.
4. Lucas,"Analysis,Design&Implementation",McGrawHill
5. JamesA.Senn,"Analysis,DesignofInformationSystems",SecondEdition,McGrawHill.

## Paper Title: Web Development/ Designing
## Paper Code: CSA-MDC-103/203/303
## (Any Semester)

| Th | Pr | Cr | |
|---|---|---|---|
| - | 3 | 3 | Practical External Marks: 65<br>Practical InternalMarks:10 |

**Level: L-100**
**Timeduration:3hours** **Number of Lectures: 90**

**Course Objectives:**

1. Identify the appropriate programming environment for developing dynamic client-side and server-side web applications.
2. Plan, develop, debug, and implement inter active client-side and server-side web applications.
3. Identify the tools needed to create dynamic client-side and server-side web applications.

### Learning Outcome:

1. Students are able to develop a dynamic webpage by the use of Java Script and HTML.

**Outlines syllabus: Web Development/Designing**

**Note:**

i. The Question Paper will consist of Four Sections.
ii. Examiner will set total of NINE questions comprising TWO questions from each Section and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Section and the Compulsory question.
iv. All questions carry equal marks unless pecified.

### Unit-I

**World Wide Web-Web Browsers –Markup Languages- Style Sheet Technologies -Client- Side, Server Side – HTML – Headings – Links-Images – Lists - Tables - Forms, Frames, Cascading style Sheets**

### Unit-II

**JAVASCRIPT-Introduction-Simple Program, Decision Making.**
**Equality and Relational Operators - Control Statements - Functions - Programmer Defined Functions, JavaScript Global Functions, Recursion**

## Unit-III

**Arrays-References and Reference Parameters, Passing Arrays to Functions, Multi dimensional Array Objects of JavaScript: Window Object - Document Object**

## Unit-IV

**Forms Object -Text Boxes and Text Areas - Buttons, Radio Buttons, Check Boxes, Select Object, Other Object: Data Object- Math Object- String Object- Arrays- Worked Examples.**

## Practical Syllabus:

**Create Web pages in HTML Links, Images, Lists, Tables, Forms, Frames, Cascading style sheets Creating Web pages in JavaScript, Simple Program, Decision Making, Equality and Relational Operators**
**- Control Statements-Functions-Programmer Defined Functions, JavaScript Global Functions Recursion-Arrays**
**Create web pages in JavaScript-Window Object-Document object-Forms- Object, Text Boxes and Text Areas, Buttons, Radio Buttons, Check Boxes, Select Object, Other Object, Data Object, Math Object, String Object, Arrays, Worked Examples**

## References:


1. Learning Mike McGrath, JavaScript, Dream, Tech Press, 1st Edition 2006

2. Jennifer Robinson, Learning Web Design, O Reilly, 2018

3. Melani and Kyrnin's "HTML, CSS, JavaScript All in One", Pearson, 2018

# Paper Title: Introduction to InformationTechnology
# Paper Code: CSA-MDC-104/204/304
# (Any Semester)

| Th | P | Cr | |
|---|---|---|---|
| - | 3 | 3 | Practical External Marks: 65<br>Practical InternalMarks:10 |
| Time duration:3hours | | | Number of Lectures: 90 |

**Objectives:**

1. To familiarize the students with the Fundamentals of InformationTechnology.
2. To assess the implications for the markets and organizational change of advances in information technologies.

**Learning Outcomes:**

- Knowledge about the basics of Windows Operating System.
- Learning various operations of Windows OS.
- Knowledge of MS word, MS PowerPoint, MS Excel, MS Access

## UNIT I

**Introduction:** Evolution of Computers, Windows and Unix. Fundamentals of windows, anatomy of windows, Operations on window: Opening, Minimizing, Maximizing, Moving, Resizing, Closing; Windows Explorer, Folders: Creating, deleting, copying, renaming folders, folder properties; Icons, Menu, Taskbar, Control panel, Recycle bin.

## UNIT II

**Word Processing Package:** Basics of Word Processing; Word Processing Basics; Text creation, Manipulation, Finding and replacing, Formatting of text; Printing of word document, Page Layout: Margin setting, Alignments, Adding Borders and shading, Adding Headers and Footers, Setting up Multiple columns, Working with tables, Spell check, Auto Correct, Grammar facility, Mail merge**.**

## UNIT III

**Spreadsheet Package:** Worksheet Basics, Data Entry in Cells: Entry of numbers, text and formulae, Moving data in a worksheet, Moving around in a worksheet, Selecting Data Range. Using the interface (Toolbars, Menus), Editing Basics, Working with workbooks, Cell referencing: Absolute, Relative and Mixed; Formatting and Calculations: using Auto fill, Working with Formulae, Creating Chart and graphs.

## UNIT IV

**Presentation Packages:** Basics, General Features, Creating a presentation, Different types of slide views, Master Slides and its use, Formatting Slides: slide design, Layout and background; Animation effect, Transition effect, timing effects, Macros.

Database Package: Introduction to Database, Tables, Data Types, Attributes, Records; Overview of MS- ACCESS Creating Database, Creating Tables, Data types, Creating forms and queries.

**References Resources:**


1. Priti Sinha and P.K. Sinha: Computer Fundamentals, BPB Publications, 2004

2. V. Rajaraman and N. Adabala: Fundamentals of Computers, PHI, 2014

3. R. K. Texali: PC Software made simple, McGraw Hill., 2017

## Computer Programming and Problem Solving

## Paper Code: CSA-MDC-105/205/305

## (Any Semester)

| Th | P | Cr | Theory External Marks: 65 |
|---|---|---|---|
| 3 | - | 3 | Theory Internal Marks:10 |

**Level: L-100**
**Time duration:3hours**
**Number of Lectures: 90**

**Course objectives:**

- To introduce students to the foundations of computing, programming and problem-solving.
- To develop basic programming skills.

**Learning Outcomes:**

- Students will be able to break down real-world problems into smaller, manageable components.
- Students will be able to develop an ability to represent problem solutions using algorithms, flowcharts, and pseudocode.
- Students will be able to implement the algorithms using suitable programming languages.
- Students will be able to write, test, and debug simple programs.

**Prerequisites:** No pre-requisites required, suitable also for those with school level programming skills.

**UNIT - I**

1. **Introduction to problem solving using computers.**
   Problem solving steps**,** What are programs, Computing based applications of programming, Steps followed in Program Development, Good Programs and Bad Programs, Programming Languages as tools, types of languages (machine level, assembly, high level), programming paradigms (Procedural, Object oriented, Database processing, Scripting)

2. **Introduction to Basic Computing Tools**: Algorithms - definition, characteristics, examples, advantages and limitations**; Flowcharts** - definition, notations, examples, advantages and limitations; **Pseudo codes** - notations, examples, advantages and limitations; **Decision Tables**-definition, advantages, limitations, examples; **comparison** between algorithms, flowchart and pseudo codes.

**UNIT - II**

3. **Converting pseudo-code to programs**: Compilation process (compilers , interpreters), linking and loading, syntax and semantic errors, testing a program, Good Programming Practices (naming conventions , documentation, indentation), Difference between algorithms and programs

4. **Basic Elements of a Program**: Identifiers, keywords, notion of data types, variables and constants, statements, operators, syntax, program execution environment: Compilation/interpretation, debugging and testing, types of errors, source program, object program

**UNIT – III**

5. **Sequential and Branching statements:** Sequential statements, Branching: conditional (if, if..else, if..else ladder, nested if), unconditional (goto)

6. **Iterative statements:** what is iteration, iteration body, initialization, termination condition, iteration step, Loop controls: while do, do..while, for.

**UNIT – IV**

7. **Aspects of loop design**: Avoiding infinite loops, comparison between while do, do while and for loops, Loop control flow statements: break and continue

8. **Functions:** introduction, advantages, defining, declaring and calling a function

**Note:**

Illustration of basic programming concepts like tokens, data types, identifiers, input/ output statement, arithmetic operations, conditional statements, loops etc. using any programming language such as C, C++, Python.

**References:**

1. Subhashis Banerjee, S. Arun-Kumar, D. Dubhashi: Introduction to Computer Science.

2. Structure and Interpretation of Computer Programs by Harold Abelson and Gerald Sussman with Julie Sussman, MIT Press, 1985.

3. How to solve it by Computer by R. J. Dromey, Prentice-Hall India EEE Series.

## Paper Title: Data Analysis using Excel
## Paper Code: CSA-MDC-106/206/306
## (Any Semester)

| Th | Pr | Cr | Practical External Marks: 65 |
|---|---|---|---|
| - | 3 | 3 | Practical InternalMarks:10 |

**Level: L-100**
**Time duration:3hours**
**Number of Lectures: 90**

**Course objectives:**

- Learn to handle data, clean the data to make meaningful dataset without missing value using various excel functions.
- Learn to use excel functions to get statistical and financial results.
- Learn to use intelligent functions in excel like lookups and conditional calculations.
- Learn to visualize data, into the form of Graphs, Pivots and list with subtotals. Explore various types of graphs and chart styles.
- Equip with skills in managing and preparing data for analysis in Excel.
- Employ advanced data analysis techniques such as what-if analysis and macros.

**Learning Outcomes:**

- Collect data from different sources and organize it to a meaningful tabular format.
- Apply formulas to the data using excel built-in functions.
- Utilize Excel's advanced features for data manipulation and analysis.
- Select graph types and chart styles which will suit the kind of data they analyse.
- Explore the immense possibilities of pivot table and make meaningful reports.
- Utilize Excel's advanced features for data manipulation and analysis.

### UNIT I

**Introduction to Excel Basics:** An Overview of the Screen, Navigation and Basic Spreadsheet Concept, Various Selection Techniques, Data Entry: creating, Editing, Saving, Page Settings. Auto fill, formatting data in the cell, arithmetic operations, basic formulas, working with excel range and tables, Use of cell references in formulas (relative, absolute, and mixed references) and referencing cells outside the worksheet, demonstration of formatting worksheets, applying conditional formatting, Protecting and Un-Protecting worksheets and cells, Shortcut Keys
**Basic Functions:** arithmetic, string, date & time
**Statistical Functions:** Counting items in a dataset, Means, Modes and Medians, Finding Values, Ranks and Percentiles, Standard Deviations and Variances, Regression Analysis, Correlation, t-distributions

## UNIT II

**Financial functions:** PMT, FV, NPV, IRR, PV, RATE, NPER, PPMT, XNPV, XIRR

**Advanced Excel Functions for Data Analysis:** Using formulas for Conditional Analysis – IF, AND and OR functions to evaluate simple and multiple conditions; conditional calculations(SUMIF,COUNTIF, AVERAGEIF); Matching and Lookups(VLOOKUP, HLOOKUP, INDEX, MATCH).

**Data Visualization with Excel:** Demonstration of creation of a bar chart representing sales data for different months; Customize the appearance of a chart; Use of various types of charts in Excel-column charts, histograms, line charts, pie charts and scatter charts.

Use of spark lines to display trends within a single cell.

## UNIT III

**Introduction to Pivot Tables and Pivot Charts:** Creating a PivotTable using a dataset, Customizing Pivot Tables, Creating a Pivot Chart based on an existing PivotTable, Analyzing Data with PivotTables.

**Managing and AnalysingData:** Importing data into Excel from an external source, Cleaning Data, Data sorting and filtering, Using Data Validation.

**Advanced Data Analysis Techniques:** Performing Spread sheet What-If Analysis; Analysing Data Using Goal Seeking and Solver.

## UNIT IV

**Introducing Data Analysis Toolbar Tools:** Using the Descriptive Statistics tool, Creating a histogram, Ranking by percentile, Calculating moving averages, Using the Exponential Smoothing tool, Using Data Analysis t-test tools, Using the Regression and Correlation tools, Implementing the ANOVA data analysis tools.

**Excel Macros:** Recording and editing macros, Managing recorded macros.

**Reference Books**


- Excel Data Analysis Modeling and Simulation., Hector, G.(2019), Springer.
- Microsoft Excel data analysis and business modeling ,Winston, W.(2016), Microsoft press.
- Excel 2019 bible, Alexander, M., Kusleika, R., &Walkenbach, J. (2018), John Wiley & Sons.
- Excel data analysis for dummies, Nelson, S.L., & Nelson, E.C. (2014), John Wiley & Sons.
- Data Analysis with Microsoft Excel, Berk, K.N., & Carey, P. (1998),Pacific Grove, CA: Duxbury Press.

**Paper Title : Social Media Marketing**

**Paper Code : CSA-MDC-107/207/307**

**(Any Semester)**

| **Th** | **P** | **Cr** | | |
|---|---|---|---|---|
| 3 | - | 3 | **Theory External Marks** | 75 |
| | | | **Theory Internal Marks** | -- |
| **Time Duration** : 3 Hrs. | | | **No. of Theory Hours** | 45 |

**Objectives :**

- To understand the evolution, concepts, and significance of social media marketing.
- To develop skills for planning and implementing effective social media strategies.
- To understand ethical issues and responsible practices in social media marketing.
- To explore emerging trends and applications of social media across different sectors.

**Learning Outcomes:** After completing this course, students will be able to:

- Understand the concepts, evolution, and significance of social media marketing.
- Develop effective social media strategies and create engaging digital content.
- Identify and analyze the features and applications of major social media platforms.
- Recognize ethical issues and responsible practices in social media marketing.
- Evaluate emerging trends and applications of social media in different sectors.

Note:

i. The Question Paper will consist of Four Sections.
ii. Examiner will set total of NINE questions comprising TWO questions from each Section and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Section and the Compulsory question.
iv. All questions carry equal marks unless specified.

**Unit I**

**Understanding Social Media** : Evolution of Social Media, Social Media Ecosystem, Role of Social Media in Digital Marketing, Features and Characteristics of Social Media Platforms, Consumer Behaviour in Social Media, Social Media Marketing vs Traditional Marketing, Types of Social Media Platforms: Social Networking, Video Sharing, Blogging, Microblogging, Forums and Communities; Overview of Major Platforms: Facebook, Instagram, LinkedIn, X (Twitter), YouTube, Pinterest, Snapchat

**Unit II**

**Planning and Creating Social Media Strategies:** Developing Social Media Marketing Strategy, Goal Setting (SMART Goals) Identifying Target Audience. Customer Persona Development, Brand Positioning on Social Media, Content Marketing Fundamentals, Content Planning and Content Calendar, Types of Content: Images, Videos, Reels, Stories, Infographics, Blogs, Live Streaming, Copywriting for Social Media.

**Unit – III**

**Responsible and Ethical Use of Social Media:** Meaning and Importance of Ethics in Social Media Marketing, Consumer Privacy and Data Protection, Truthful Advertising and Consumer Protection, Intellectual Property Rights and Copyright Issues, Ethical Practices and Influencer Marketing, Crisis Communication and Responsible Social Media Practices.

**Unit IV**

**Future Trends and Practical Applications :**Artificial Intelligence (AI) and Automation in Social Media, Influencer Marketing and the Creator Economy, Short-form Video Content and Live Streaming, Social Commerce and In-app Shopping, User-Generated Content (UGC) and Brand Communities, Social Listening and Online Reputation Management, Applications of Social Media in: Business and Marketing, Education, Healthcare, Tourism and Hospitality, Government and Public Services

**Suggested Readings:**


- Tuten, Tracy L., & Solomon, Michael R. (2023). *Social Media Marketing*. SAGE Publications.
- Tuten, Tracy L., & Solomon, Michael R. (2018). *Social Media Marketing: Principles and Strategies*. SAGE Publications.
- Hanna, Richard K., Rohm, Andrew, & Crittenden, Victoria L. (2011). *We're All Connected: The Power of the Social Media Ecosystem*. Business Horizons / Elsevier.
- Chaffey, Dave, & Smith, P. R. (2022). *Digital Marketing Excellence: Planning, Optimizing and Integrating Online Marketing*. Routledge.
- Kingsnorth, Simon. (2022). *Digital Marketing Strategy: An Integrated Approach to Online Marketing*. Kogan Page.

**Paper Title : Business Intelligence**

**Paper Code : CSA-MDC-108/208/308**

**(Any Semester)**

| **Th** | **P** | **Cr** | | |
|---|---|---|---|---|
| 3 | - | 3 | **Theory External Marks** | 75 |
| | | | **Theory Internal Marks** | -- |
| **Time Duration** : 3 Hrs. | | | **No. of Theory Hours** | 45 |

**Objectives :**

- To understand the basic concepts, framework, and importance of Business Intelligence (BI).
- To familiarize students with BI tools and data visualization techniques.
- To develop awareness about ethical issues related to data usage and BI practices.
- To understand the applications and future trends of Business Intelligence in business decision-making.

**Learning Outcomes:** After completing this course, students will be able to:

- Understand the basic concepts, components, and processes of Business Intelligence.
- Identify the importance of data governance, privacy, security, quality, and ethical practices in Business Intelligence.
- Recognize ethical issues related to data privacy, security, accuracy, and responsible BI practices.
- Understand emerging trends and applications of Artificial Intelligence in Business Intelligence.

Note:

v. The Question Paper will consist of Four Sections.
vi. Examiner will set total of NINE questions comprising TWO questions from each Section and ONE compulsory question of short answer type covering whole syllabi.
vii. The students are required to attempt ONE question from each Section and the Compulsory question.
viii. All questions carry equal marks unless specified.

**Unit I**

**Introduction to Business Intelligence** : Meaning and Concept of Business Intelligence (BI), Importance and Role of Business Intelligence in Business Decision-Making, Evolution of Business Intelligence, Business Intelligence Components, Data as a Business Resource, Types of Business Data: Structured Data, Semi-Structured Data and Unstructured Data.

## Unit – II

**Business Intelligence Architecture and Business Analytics :**Introduction to Business Intelligence Architecture, Working Process of a Business Intelligence System : Data Collection, Data Integration, Data Storage, Data Processing and Analysis, Reporting and Decision Support. Basic Concept of Decision Support Systems (DSS) – Meaning and Applications.

Business Analytics : Introduction, Relationship between Business Intelligence and Business Analytics, Types of Business Analytics : Descriptive, Predictive and Prescriptive Analytics, Business Reporting and Performance Measurement, Key Performance Indicators (KPIs).

## Unit – III

**Data Governance, Privacy and Security in Business Intelligence:** Introduction to Data Governance in Business Intelligence, Importance of Data Governance in Organizations, Data Privacy and Protection, Data Security and Access Control, Data Accuracy, Quality, and Transparency.

Ethical Issues in Data Collection and Analysis, Bias and Fairness in Business Intelligence

## Unit IV

**Applications and Future Trends of Business Intelligence :** Applications of Business Intelligence in: Marketing and Customer Analysis, Financial Analysis, Sales Forecasting, Supply Chain Management, Human Resource Management; Role of Artificial Intelligence (AI) in Business Intelligence, Future Trends and Challenges in Business Intelligence.

**Suggested Readings:**


- Turban, Efraim, Sharda, Ramesh, Delen, Dursun, & King, David. (2021). Business Intelligence, Analytics, and Data Science: A Managerial Perspective. Pearson Education.
- Sharda, Ramesh, Delen, Dursun, & Turban, Efraim. (2024). Business Intelligence, Analytics, and Data Science: A Managerial Perspective. Pearson.
- Loshin, David. (2013). *Business Intelligence:* The Savvy Manager's Guide. Morgan Kaufmann, Elsevier.
- Kimball, Ralph, & Ross, Margy. (2013). The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling. Wiley.
- Provost, Foster, & Fawcett, Tom. (2013). Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking. O'Reilly Media.


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